

The empirical recurrence for OEIS A233712: recovering a conjecture that is not written down

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Abstract

OEIS A233712 records “Empirical recurrence of order 75”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on $S = 625$ vertices and satisfies some monic recurrence of order at most S ; Berlekamp–Massey applied to $2S$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 75, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 75 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A233712 is “Number of $(n+1) \times (3+1)$ 0..4 arrays with every 2×2 subblock having the sum of the squares of the edge differences equal to 14.”. It has offset 1 and begins

1992, 16004, 136960, 1291820, 12203480, 117736088, 1132387344, 10952782784, ...

The entry states:

Empirical recurrence of order 75 (see link above)

As of the “Last modified” line on the live entry (Jun 02 2025 08:57:46, revision 7) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 625$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S = 625$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 1250$ exact terms of the model returns order 75 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{75} c_i a(n-i),$$

c_1	2	c_{20}	−3829932326938807	c_{39}	−5719388252486897950308	c_{58}	36473104504
c_2	267	c_{21}	6644096406478322	c_{40}	−4804274071498855060830	c_{59}	−30434888090
c_3	−452	c_{22}	33620412536163789	c_{41}	11570733419549562254010	c_{60}	−6484999182
c_4	−31149	c_{23}	−59957629785518266	c_{42}	7855369665305681813210	c_{61}	99530072607
c_5	48608	c_{24}	−246310925916316094	c_{43}	−19924478037734085537194	c_{62}	1068678379
c_6	2158423	c_{25}	451573015065807790	c_{44}	−10822357171676488755857	c_{63}	−2564995415
c_7	−3263100	c_{26}	1514941619676922221	c_{45}	29148016174093418658714	c_{64}	3241218676
c_8	−100962722	c_{27}	−2855995410664906532	c_{46}	12498693996080285937608	c_{65}	5089603878
c_9	152135886	c_{28}	−7857878356517300912	c_{47}	−36128903485128416851976	c_{66}	−104678610
c_{10}	3420174268	c_{29}	15242460624624030554	c_{48}	−12016381112916299850889	c_{67}	−753467503
c_{11}	−5218583970	c_{30}	34487752224832727449	c_{49}	37806475879943865780592	c_{68}	1766672479
c_{12}	−87698245409	c_{31}	−68907928299386098162	c_{50}	9526613331603464874991	c_{69}	795611864
c_{13}	136641649548	c_{32}	−128372317874920304136	c_{51}	−33246879493213077895748	c_{70}	−17518401
c_{14}	1754186167962	c_{33}	264624141671456684474	c_{52}	−6146346961361799697929	c_{71}	−55902753
c_{15}	−2802711220100	c_{34}	405789151671720069833	c_{53}	24429394234489194825078	c_{72}	95438418
c_{16}	−27971827456828	c_{35}	−864940381145970754452	c_{54}	3164752775333462071278	c_{73}	231106918
c_{17}	45913611557018	c_{36}	−1089744190919702160629	c_{55}	−14892010343738297656716	c_{74}	−2189260
c_{18}	361378448931754	c_{37}	2409006394110880654884	c_{56}	−1259911846717588907992	c_{75}	−4197931
c_{19}	−609786115736312	c_{38}	2484935877347857881531	c_{57}	7465057802284954905920		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 75, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $75 + 4$ terms removed returns order 75 as well, so $\deg q_{\min} = 75 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 16 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $75 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 75 that this sequence satisfies — and that family turns out to have exactly one member.

References

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